Remark on general sol'n to PDE:

For <u>ODE</u>: 1st-order linear combination of one sol'n

2nd-order " " " two sol'ns

3rd-order " " " three sol'ns

For 1st-order PDE: (2 indep. variables u_1, u_2) general sol'n is a f'n of 2 $f(u_1, u_2)$ linearly indep. sol'ns

For PDE, take for example problem 6 on exam. Sep. of Variables $\Rightarrow$

$X'' + \mu X = 0 \qquad Y' + \mu Y = 0 \qquad (u_{xx} - u_y = 0)$

$X(x)$ = linear combination of $\sin(\sqrt{\mu}\, x)$, $\cos(\sqrt{\mu}\, x)$

$Y(y)$: $e^{-\mu y}$

$e^{-\mu y}\sin(\sqrt{\mu}\, x)$, $e^{-\mu y}\cos(\sqrt{\mu}\, x)$ if on a disk $\sqrt{\mu} = n$

Fourier series: linear combinations

HW ch 5 §9 # 9.2 $u(x) = \int_I g(\lambda) u(x,\lambda)\, d\lambda$ is a sol'n to nonhomog. eq'n.

HW ch 7 §2 # 2.7 $e^{-\lambda y}\cos(\lambda x)$, $e^{-\lambda y}\sin(\lambda x)$

Showed that $\int_0^\infty e^{-\lambda y}\cos(\lambda x)\, dx$ is also a sol'n

Review Ch 8 §5 sol'n to IVP for wave eq'n, $n=2,3$

If we can solve

$$\Delta v_\rho - (v_\rho)_{tt} = 0 \qquad \underline{x} \in \mathbb{R}^n,\ t>0$$

$$\left.\begin{aligned} v_\rho(\underline{x},0) &= 0 \\ \frac{\partial v_\rho}{\partial t}(\underline{x},0) &= \rho(\underline{x}) \end{aligned}\right\} \underline{x} \in \mathbb{R}^n$$

$\leftarrow$ includes $t=0$

where $v_\rho \in C^3(\mathbb{R}^n) \cap C^3(\overline{\mathbb{R}^+})$

Then $u = \frac{\partial v_\rho}{\partial t}$ satisfies

$$\Delta u - u_{tt} = 0 \qquad \underline{x} \in \mathbb{R}^n,\ t>0$$

$$\left.\begin{aligned} u(\underline{x},0) &= \rho(\underline{x}) \\ u_t(\underline{x},0) &= 0 \end{aligned}\right\} \underline{x} \in \mathbb{R}^n$$

$u(\underline{x},0)$ $u_t(\underline{x},0)$

So let u_ψ be the sol'n with $0, \psi$ data

u_ϕ " " " " $0, \phi$ data

Then $(u_\phi)_t$ is " " " $(\phi, 0)$ data

and by superposition $u = u_\psi + \frac{\partial}{\partial t}(u_\phi)$ is the sol'n with (ϕ, ψ) data.

IVP for wave eq'n

$$n=3 \quad u(\underline{x},t) = \frac{1}{4\pi t}\int_{S(\underline{x},t)} \psi(\underline{x}')\,d\sigma_t + \frac{\partial}{\partial t}\left[\frac{1}{4\pi t}\int_{S(\underline{x},t)} \phi(\underline{x}')\,d\sigma_t\right]$$

boundary of Ball in 3D.

$$n=2 \quad u(\underline{x},t) = \frac{1}{2\pi}\int_{\overline{B}(\underline{x},t)} \frac{\psi(x_1',x_2')\,dx_1'dx_2'}{\sqrt{t^2-(x_1-x_1')^2-(x_2-x_2')^2}}$$

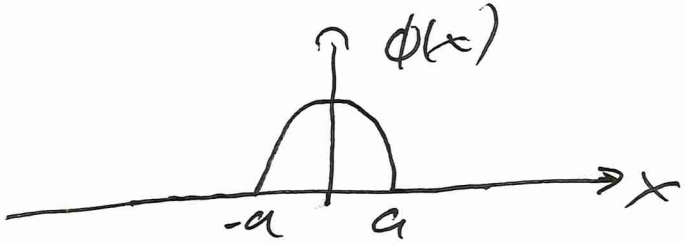
$$+\frac{\partial}{\partial t}\left\{\frac{1}{2\pi}\int_{\overline{B}(\underline{x},t)} \frac{\phi(x_1',x_2')\,dx_1'dx_2'}{\sqrt{}}\right\}$$

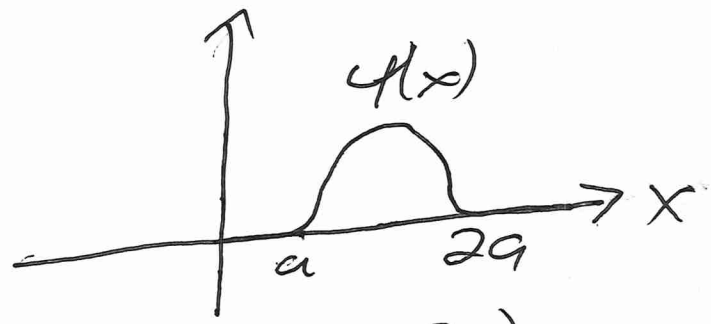
The entire disk

$ds \to \delta(x,y)\,dA$

Ch 8 S6 Huygen's Principle

Suppose the <u>support of ϕ + ψ</u> (~~the~~ an interval that contains all the places where both ϕ + ψ are not zero)

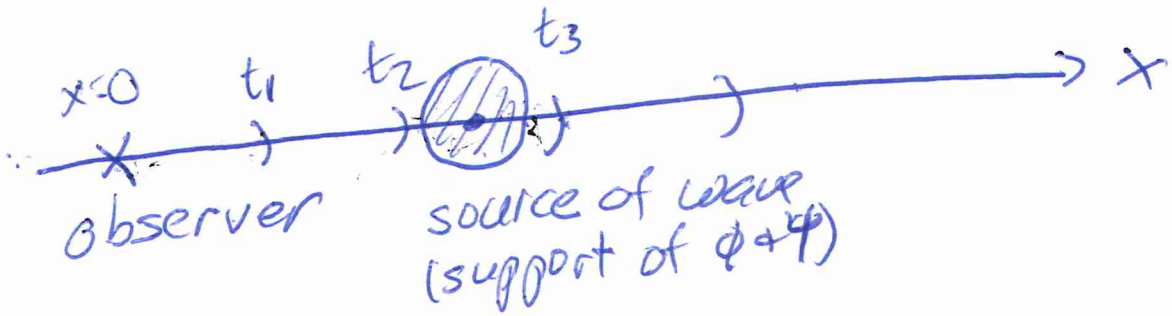




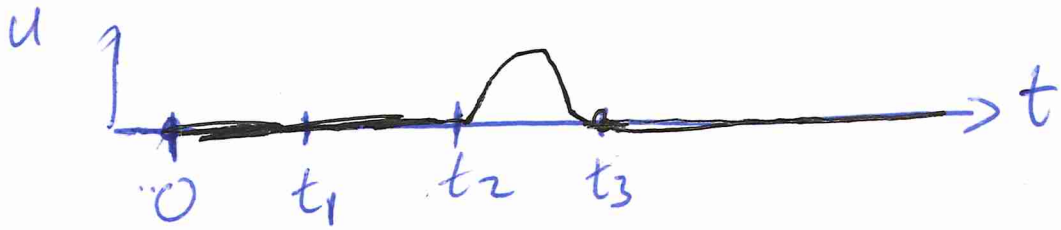
supp $\{\phi + \psi\} = (-a, 2a)$ or $(-2a, 3a)$

so outside the interval, ~~bot~~ $\phi(x) = \psi(x) = 0$.

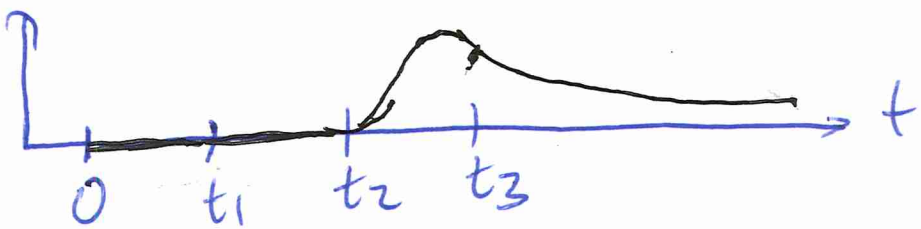
is small.



$n = \cancel{2}\,3$



<u>$n=2$</u>



<u>Huygen's principle</u> $u(\underline{x},t)$ depends only on data (and a deriv) over the bdry $S(\underline{x},t)$ for

$$n = 3, 5, 7, 9, \ldots$$

(odd dimensional spaces, sound come & go).

$n=1$? $$u(x,t) = \underbrace{\tfrac{1}{2}\left[\phi(x+t) + \phi(x-t)\right]}_{\text{like } n=3} + \underbrace{\tfrac{1}{2}\int_{x-t}^{x+t} \psi(x')\,dx'}_{\text{like } n=3}$$

<u>Note</u> For all Thms thus far in ch 8, we have assumed (at least) that $\phi \in C^2(\mathbb{R})$ and $\psi \in C^1(\mathbb{R})$ so that $u \in C^2$.

What if ϕ (or ψ) is only piecewise continuous?

Then u would only be sectionally continuous & how would we make sense of $u_{xx} = u_{tt}$?

This gets us to "generalized sol'n" involving "distributions" (generalization of f'n)

ex) $\delta(x)$ $\quad \frac{d}{dx}[H(x)] = \delta(x)$

But from a practical stand pt, we can just continue using our formulas.

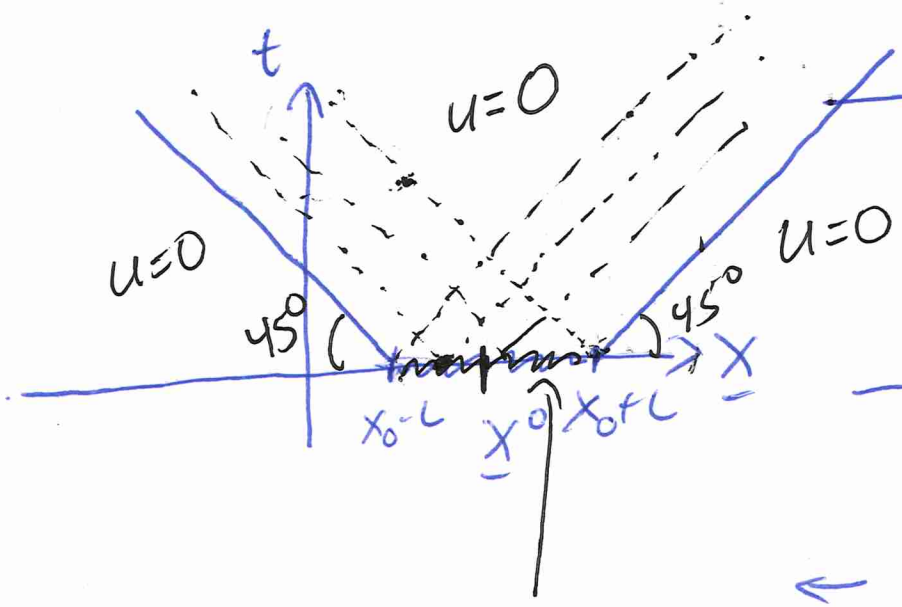
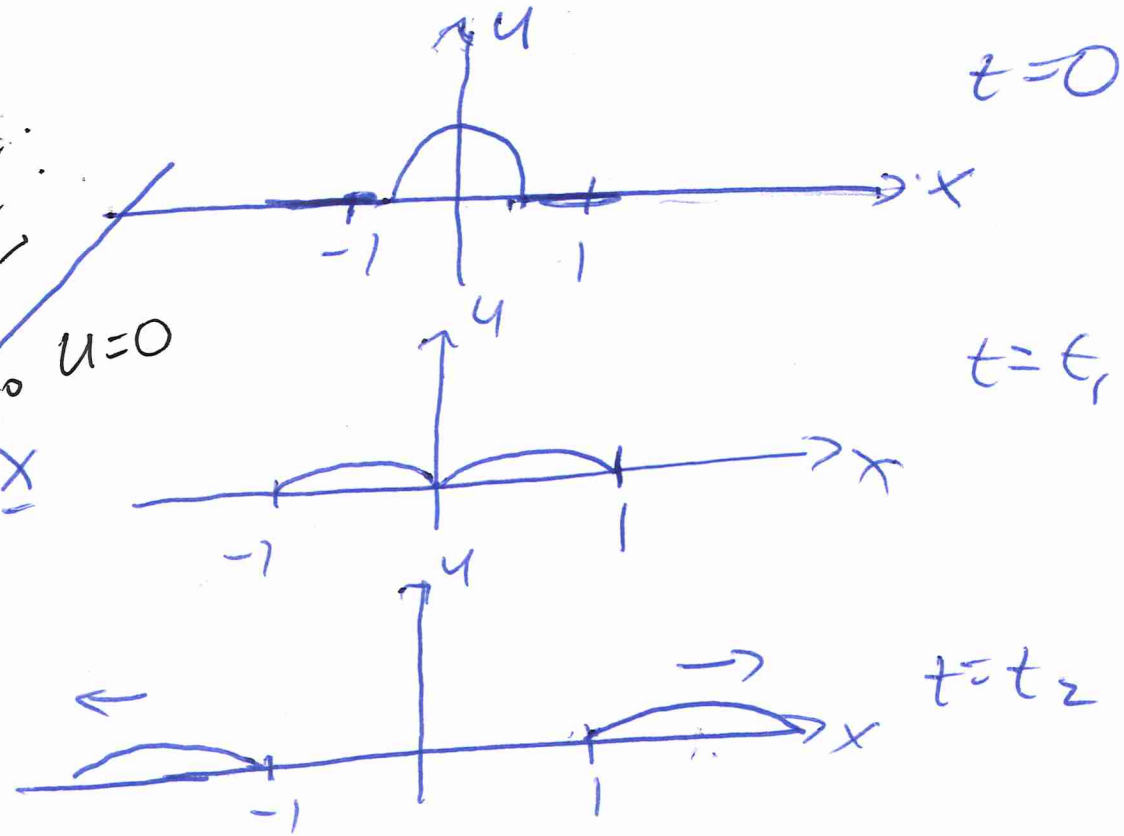
We will see an example in 2 more pp.

Recall $\phi(x) = \begin{cases} \cos(\pi x) & |x| \le \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$

$n=1, \quad \psi(x) = 0$

$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right]$

what happens spherically $n=3$



support for $\phi + \psi$ in 3 dim.